PICOPATT progress update

Simulated picoclimate testing, stability, and baseline expansion

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Scope reminder

Current data status

I still do not have the real PICOPATT data. The picoclimate work in the repository is a simulated test fixture used for validation only.

- Simulated picoclimate fixture: 4 timesteps, about 23 variables, used to test the pipeline structure.
- Real PICOPATT validation remains future work once the real data is available.
- Current focus: keep the explanation format stable and broaden the benchmark matrix.

What changed recently

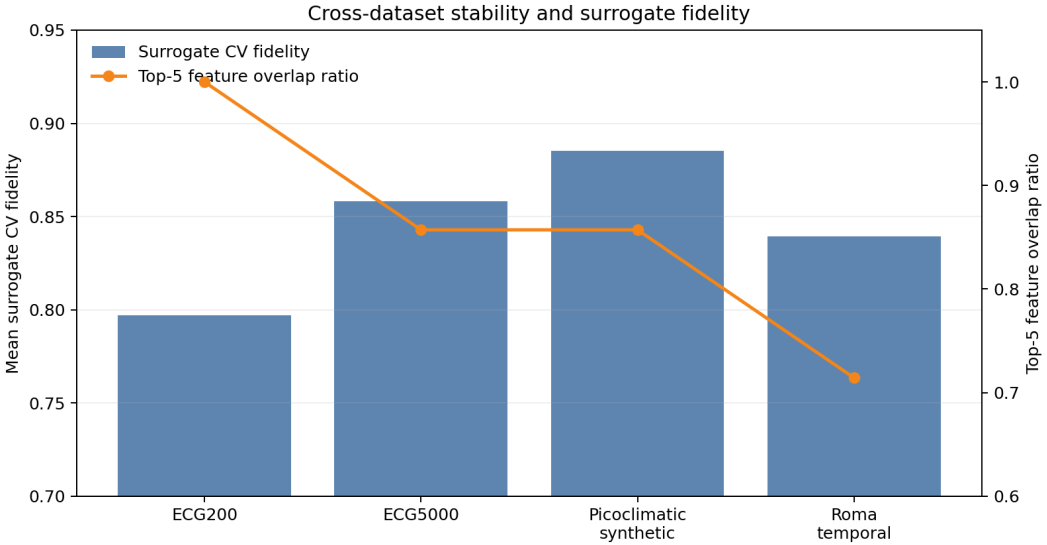
Validated in code

- Shapelet notebooks renamed and aligned under `shapelets_`
- Cross-dataset stability run completed with 7 seeds
- HDBSCAN added to the picoclimate benchmark script
- Autoencoder + KMeans baseline already available and runnable

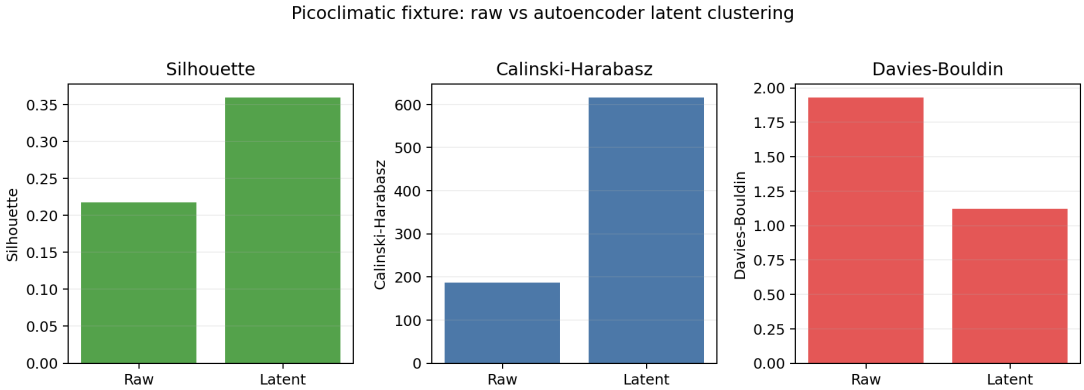
Governance update

- Explanation output format is now treated as a contract
- Future methods must emit the same cluster-summary structure
- The goal is sensor-readable explanations, not feature-index-only narratives

Evidence: stability across datasets

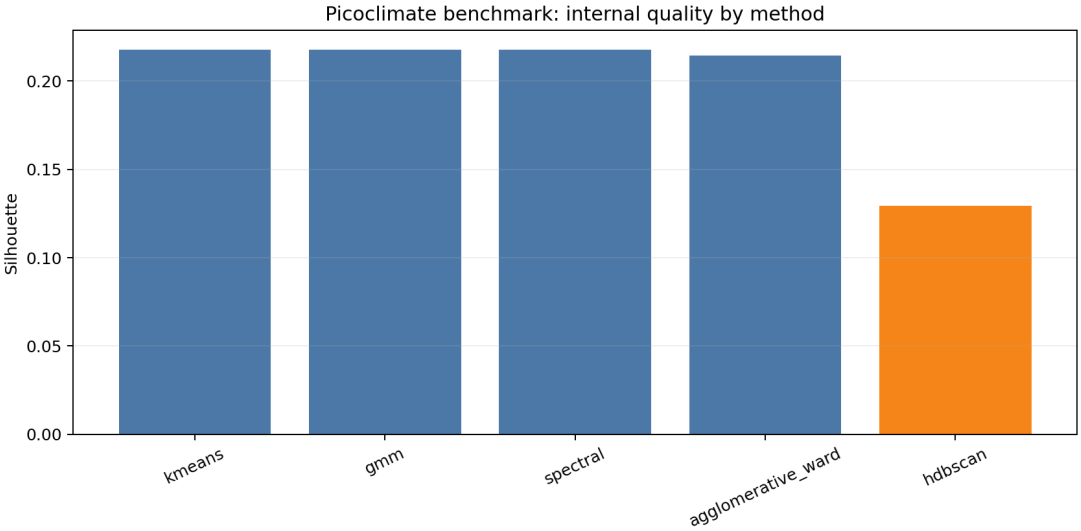


Evidence: deep representation baseline



- The autoencoder baseline already existed in the repository and is now validated on the simulated picoclimate fixture.
- Latent space improves clustering quality over the raw feature baseline in this quick test.
- This gives a concrete representation-space comparator for the next phase.

Evidence: spatial baseline expansion



Interpretation layer

What is already supported

- Surrogate fidelity is high enough to defend model-faithful explanations.
- SHAP concentration already gives a compact narrative for the shapelet baseline.
- Stability is now measured rather than assumed.

What is still missing

- Sensor-unit explanation template frozen across methods
- Spatial density comparators on real PICOPATT data
- A fuller joint-optimization method if the paper goes beyond a pipeline story

Recommended next phase

1. Freeze the explanation output format and make every method emit the same cluster-summary schema.
2. Keep ExKMC as the main interpretable clustering baseline for now.
3. Extend the benchmark matrix with spatial methods; HDBSCAN is the first one now in place.
4. Treat the autoencoder + KMeans path as the deep representation baseline.
5. If the research contribution shifts toward methods, prototype the simplified ExKMC-style spatial regularizer next.

Practical takeaway

The current repository is now ready for a Phase 2 story: simulated validation, stable explanations, and broader clustering baselines before moving to real PICOPATT data.

Slide space for future figures

Reserved space

Reserved for future PICOPATT cluster visualizations, sensor-readable summaries, or spatial maps.

Suggested future figures: cluster profile heatmap, station map, and one explanation example with sensor units.